

Quantum Mechanics III

Lecture 11

Parameter Estimation and Experimental Design

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

■ Operational setting

Preparation and measurement procedures are distinguished from their matrix representations.

$$F_{ij} = E[\partial_i \ln p \partial_j \ln p]$$

■ Mathematical representation

Positivity, trace, and tensor-product conventions are fixed before the main calculation.

$$F_Q = \text{Tr}(\rho L_i L_j)$$

■ Dynamics and measurement

The map from a state to observed probabilities is written explicitly for the experiment under discussion.

$$x_{next} = \operatorname{argmax}_x E_D[KL(p(\theta|D, x)||p(\theta|D))]$$

■ Consistency conditions

Complete positivity, normalization, Hermiticity, and appropriate classical limits are checked as needed.

■ Example

The example is worked for a qubit or two-mode system before extending the notation to higher dimension.

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Reading: Park, ch. 11; QDC design profile; State Reconstruction Service design guide